Running CFD in batch mode

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1. Purpose

This document describes a guideline to launch CFD analysis on high computing servers (HPC).

Based on the background of the company some examples with ANSYS CFX and FLUENT code settings are described.

2. Scope

This procedure must be intended as a guideline to run a CFD simulation in batch mode.

The proposed setting will guarantee an optimization of the simulation time.

The procedure is applicable to all CFD calculations.

The procedure is applicable for the Nova Milanese and La Spezia plants

3. Definitions

CFD computation fluid dynamic

GUI graphical interface

HPC High performances computer

OS Operative System

PBS Portable Batch System [1]

4. General Procedure - Description

CFD analyses require computational time to iterate. This process is repeated whenever some differences on the CFD model must be solved. During a typical design phase, several

modifications of the CFD model must be studied (due to geometry, physic condition, or numerical setting). In addition to that, depending on the complexity of the model to be analysed by means of CFD tools, huge computational grid should be used (several Milions of Elements), affecting the computational time.

Batch mode is a largely used method to launch CFD analysed sequentially. During batch mode no extra computational effort is deployed apart from solver activities (i.e. GUI), and this help on speed up the computational process.

Batch mode can be used to run iteratively pre-processing, solver and post processing activities

The main target of batch mode setting is to automate repetitive actions, letting the user to save time during the design process.

Even if it's possible to use the command on pre/post phase, the batch mode running is largely used mainly for the solver phase

4.1 BATCH EXECUTION COMMAND

It consists of the execution of a batch file on a prompt terminal. Depending on the installed OS and on the queue scheduler that is installed on the HPC, the main command line to run a batch file are listed below:

./<batch_filename.sh> on LINUX terminal

```
./batch_nomefile
```

<batch_filename.bat>On WINDOWS prompt terminal

```
batch_nomefle.dat
```

qsub <batch_filename> if PBS scheduler [1] is used

```
qsub batch_nomefile
```

4.2 BATCH FILES ARCHITECTURE

Batch files are slightly different, depending on each OS or scheduler. Batch files must contains a list of command lines that must be used either on pre-processing , solver and post processing phase of CFD analyses.

Batch file on Windows OS must have *.bat* suffix

Batch file on Linux OS must have *.sh* suffix

Using batch file it's possible to

- Run executable command lines
- Create loop with *for* or *if* or *while* cycles
- Set variables
- Create input/output architecures

Each batch file can handle loop of command lines by using FOR loop.

- On WINDOWS OS, the syntax to be used is listed below

  FOR /L %%A IN (1,1,200) DO (

  Command_line

)

Where:

%%A is the index, start 1, increment step 1, stop 200.

- On LINUX OS, the syntax to be used is listed below

  for i in {1..200..2};

do Command_line;

done

Where:

i is the index, start 1, increment step 2, stop 200.

To set a variable for batch file is different depending on the OS used in the machine

- For WINDOWS OS

  Set myvar = *xy* give to the variable *myvar* the *yy* value

- For LINUX OS

  myvar=xy give to the variable *myvar* the *yy* value

For sake of simplicity no more batch commands will be explained on this document. For further investigation refer to [2] or [3]

An example of batch file for different environments is shown below

Batch files on Figure 1 and Figure 2 refer to the launch of a CFX solver on local machine, parallelized on 12 cores

```
@echo off
set variable=solver.exe_path (i.e. C:\Program Files\ANSYS Inc\v232\CFX\bin\cfx5solve.exe)


"%variable%" -options (i.e. -par-local -partition 12 -size-part 2.0 -size 2.0 -def def_file.def)
```

Figure 1. <nomefile>.bat batch file for WINDOWS OS

```
#!/bin/bash

variable="solver.exe_path" (i.e."etc\ANSYS Inc\v232\CFX\bin\cfx5solve")

"%variable%" -options (i.e. -par-local -partition 12 -size-part 2.0 -size 2.0 -def def_file.def)
```

Figure 2 <nomefile> .sh batch file for LINUX OS

```
#!/bin/bash

#PBS -l select=9:ncpus=16:mpiprocs=16
#PBS -N STAGE2
#PBS -q workq
#PBS -l anshpc_pack=3

variable="solver.exe_path" (i.e."etc\ANSYS Inc\v232\CFX\bin\cfx5solve")

MACHINELIST=$(uniq -c $PBS_NODEFILE | awk '{printf "%s*%s,", $2, $1}' | sed 's/,$//')

"%variable%" -options (i.e. -par-dist $MACHINELIST -size-part 2.0 -size 2.0 -def def_file.def)
```

Figure 3 <nomefile> batch file for PBS scheduler

On Figure 3 the PBS setting to launch a CFX solver on 144 cores is shown.

#PBS -l specifies the number of nodes and the number of CPUs (i.e. 9nodes, each one with 16 cores)

#PBS -N defines the name of the job (i.e. STAGE2)

#PBS -q defines the queue to be used (i.e. workq)

#PBS -l specifies the licenses for the parallelization (i.e. anshpc_pack=4)

MACHINELIST is the variable to define the list of the cores used during the parallelization. It can change depending on the line format request by each solver.

The following command lines are used for 132 core setup using 3 ANSYS hpc_pack

#PBS -l select=8:ncpus=16:mpiprocs=16+1:ncpus=4:mpiprocs=4

#PBS -l anshpc_pack=3

4.3 Solver options for batch mode running

For sake of brevity only solver main commands in batch mode will be described.

Pre/post processing batch mode command requires the setting of special files that must be declared as input option on the command line (i.e. ccl files for CFX, jou files for FLUENT)

During batch mode each command used to launch a CFD analysis must be specified on the line command my means of options.

Typical options that can be declared during the batch mode are:

- 2d or 3d model (depending on the CFD solver)
- Double precision solver
- Number of parallel cores
- Parallel distribution mode (MPI intel, etc)
- Input file
- Output file
- Memory specification for partitioner and solver
- Initialization or interpolation mode

For Ansys CFD products more details can be found on [4]

5. Limitations

The batch mode setting described in the document should be used whenever no GUI is necessary to the user during the CFD activity

Batch mode on pre/post processing phase is a strong tool when repetitive actions are required (i.e. extract same output from several simulation). The use of batch mode for single check is not recommended.

Batch mode for solver process is recommended whenever the calculation is launched on HPC server, that is typically configured to have no GUI interface. Single CFD simulation of small 3D model should be launched with GUI interface with no batch setting, to have a prompt check of the outcomes.

6. References

- [1] J.P.Jones , PBS Pro 5.1 User Guide , 09-05-2001
- [2] https://www.gnu.org/software/bash/manual/bash.html
- [3] https://en.wikibooks.org/wiki/Windows_Batch_Scripting
- [4] https://ansyshelp.ansys.com/

7. ANNEX

7.1. Batch mode of CFX post/processing

The procedure to run a post processing analysis in batch mode using ANSYS CFX code is proposed below

Step 1: prepare a .cst file on CFD post

The cst files contains all the steps performed during the generation of a personal post processing list (i.e. generation of expressions, variables, contour, vector or other plots or charts).

It can be saved as shown in Figure 4

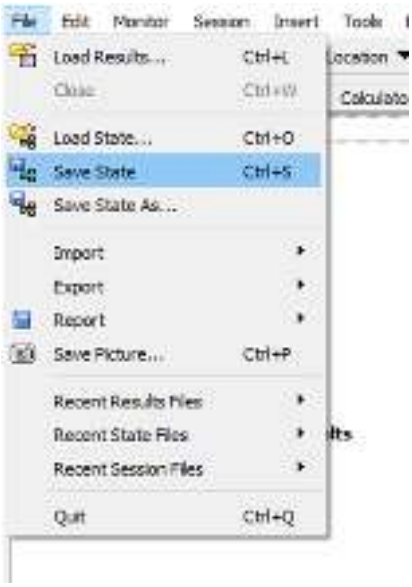

Figure 4. state file creation on CFX post

Step 2: prepare a session file (*.cse)

Session file allows to apply the settings saved on the .cst file for different cfx post results. It contains all the command lines executed during the post processing phase.

Usually, a new session cse file is recorded on the first time by means of the instructions shown in Figure 5 and Figure 6

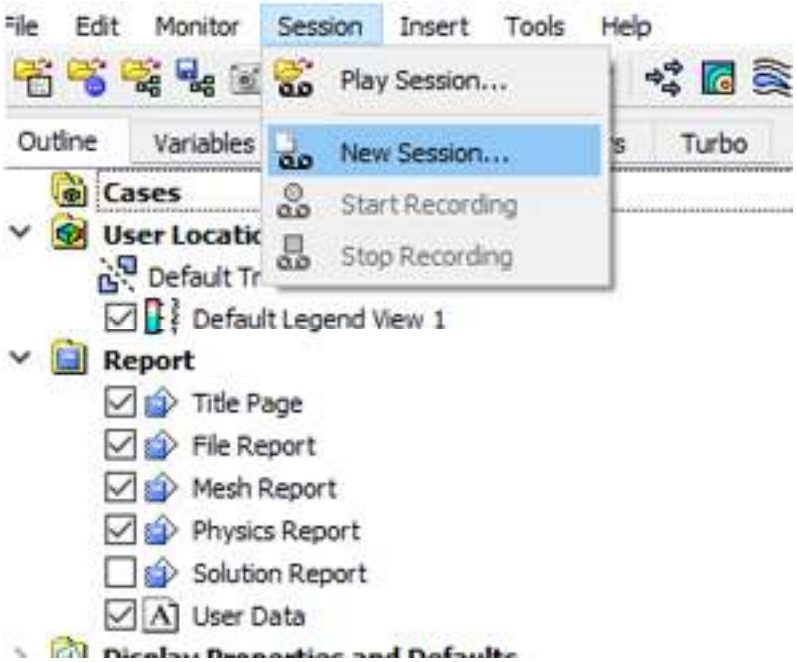

Figure 5. creation of a cse session file on CFX post

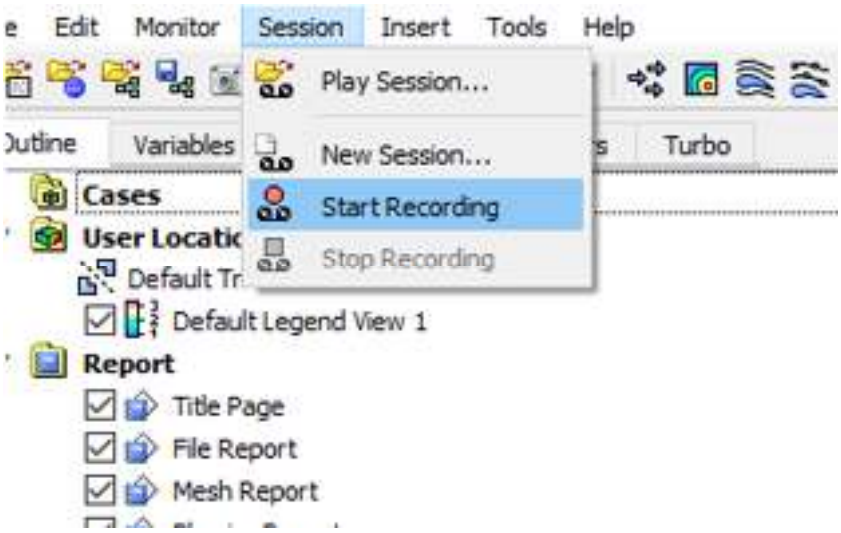

Figure 6. recording of a cse session file on CFX post

During the session recording the cse load file will be open, in order to recall all the post processing actions previously saved.

A list of typical post processing actions is

- Open a result file
- Upload a status file (.cst)
- Save a report

When all the post processing commands end, the session file will be stopped and a saved cse file is available for reuse.

To run the post processing in batch mode a batch_file similar to the one shown in Figure 7 must be launched

```
cfd5post.exe -batch <nome_file.cse> <nome_file>.res
```

Figure 7. batch command line for CFX post

7.2. Batch mode of CFX for solver

The procedure to run a solver analysis in batch mode using ANSYS CFX code is proposed below.

Typically 3 simulations are performed during a CFD analysis of centrifugal pump:

- Initial, steady state analysis, that require a definition file (*.def) to be prepared.
- Unsteady state analysis, that requires a result file must be interpolate into the mesh to initialize the unsteady simulation.
- NPSH analysis, where the pressure drop simulation campaign is usually executed. This requires the next analysis to start using the results of the previous operative pressure conditions.

For each of the simulation above-described different setting of the CFD batch must be used as shown in Figure 8, Figure 9 and Figure 10.

```
"%solve%" -par-local -partition 12 -size-part 2.0 -size 2.0 -def OP2.def
```

Figure 8. example of CFX solver launch in batch mode of first, steady state analysis - WINDOWS OS

```
"%solve%" -par-local -partition 12 -size-part 2.0 -size 2.0 -def OP3.def -ini-file OP2_001.res
```

Figure 9 example of CFX solver launch in batch mode for unsteady state initialization – WINDOWS OS

```
"%solve%" -par-local -partition 12 -size-part 2.0 -size 2.0 -def OP3.def -cont-from-file OP2_001.res
```

Figure 10. example of CFX solver launch in batch mode for NPSH analysis – WINDOWS OS

7.3. Batch mode of Ansys Fluent for solver

To launch fluent code in batch mode is necessary to create a journal file, where all the command lines used to run the simulation are defined.

And example of Fluent journal file is shown in Figure 11

```
; Read case file
rc example.cas
; Initialize the solution
/solve/initialize/hyb-initialize
; Calculate 50 iterations
it 50
; Write data file
wd example.dat
; Write solution as Ansys CFD
/wr cfd
; Write solution data file
wd example.dat
; Exit fluent
exit
y/n
```

Figure 11. example of journal file

One example of batch file for launching Fluent is shown in Figure 12. It describes a fluent calculation with 3d model, double precision mode, parallelized on 12 cores and starting from journal_file.

-g option: run Fluent without GUI

-gu option: run Fluent in a minimized GUI (useful for remote desktop access)

```
"%solve%" 3ddp -t 12 -g -i journal_file
```

Figure 12. example of FLUENT solver launch in batch mode – WINDOWS OS